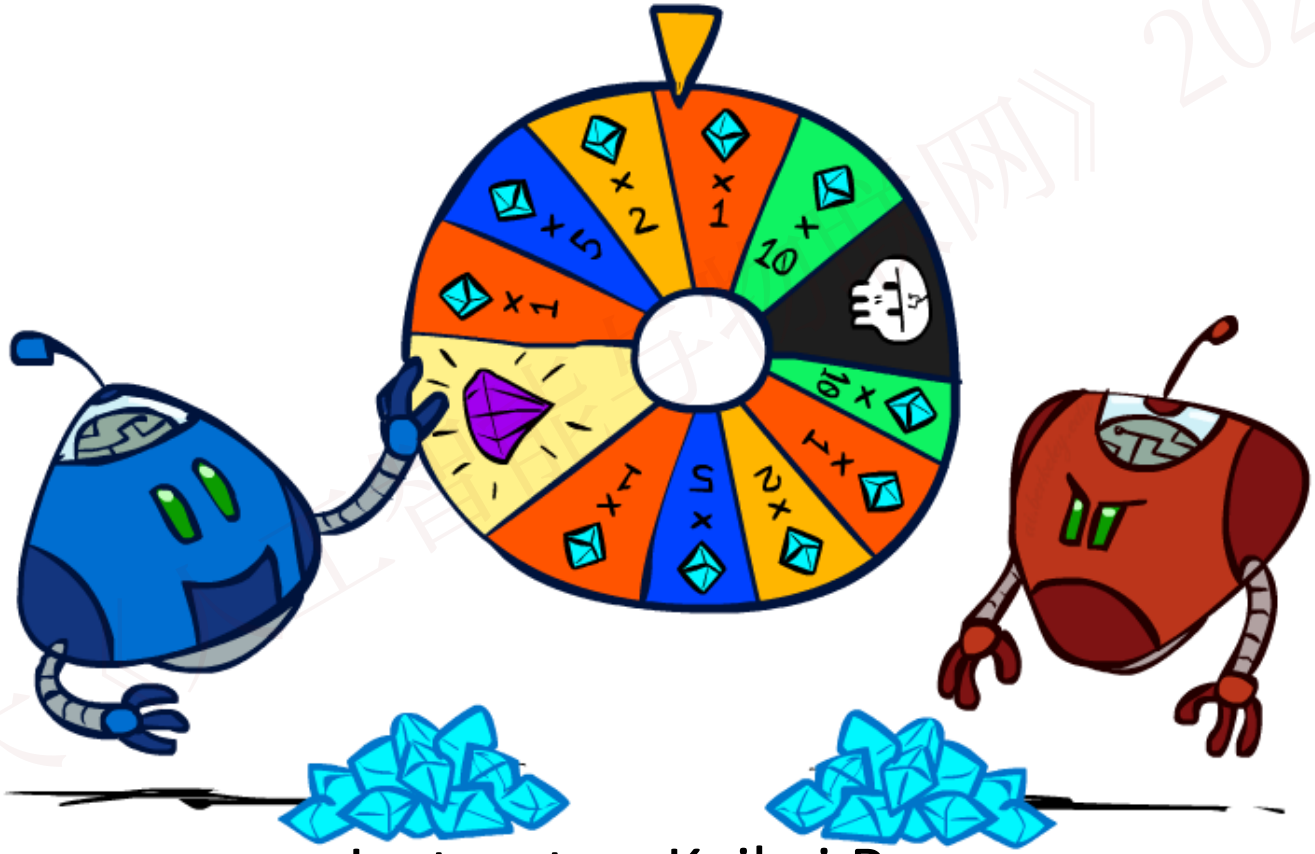
Artificial Intelligence & IoT

Expectimax Search & Utility

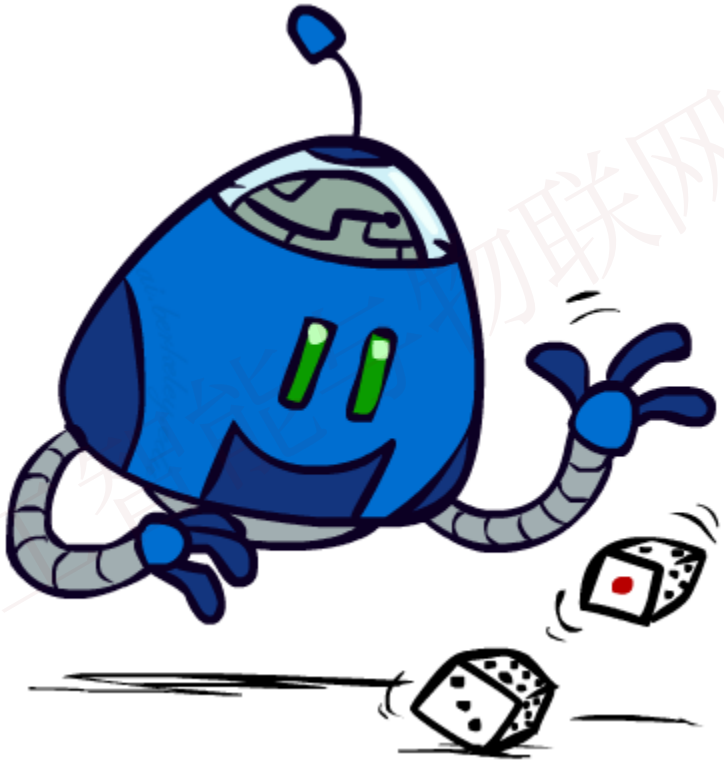


Instructor: Kaikai Pan

Zhejiang University

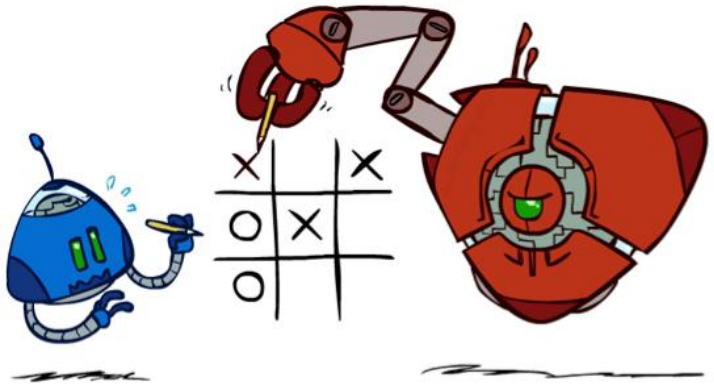
[These slides were adapted from the slides by Dan Klein and Pieter Abbeel for CS188 Intro to AI at UC Berkeley. http://ai.berkeley.edu.]

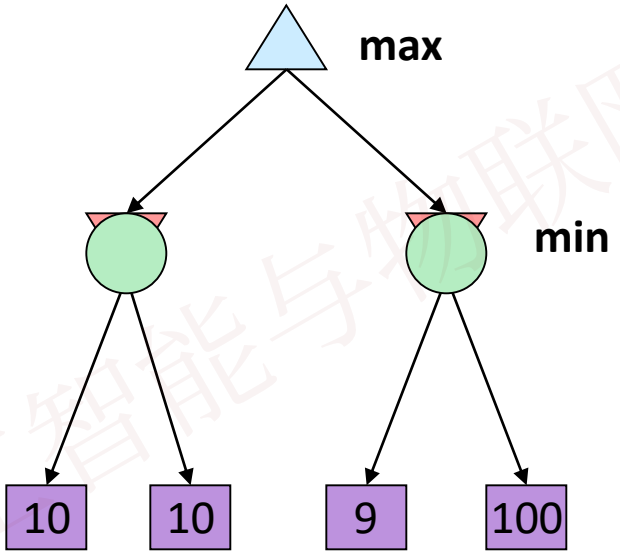
Uncertain Outcomes

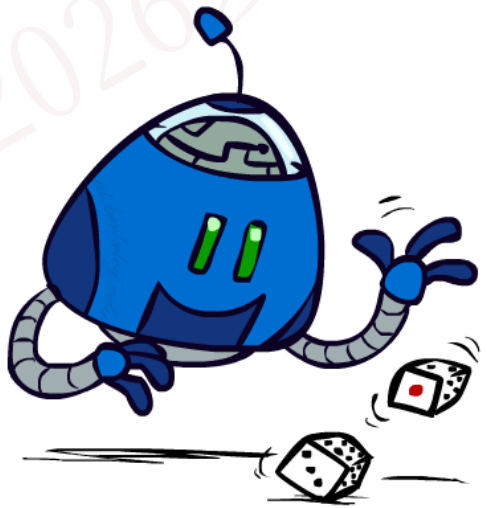


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Worst-Case vs. Average Case





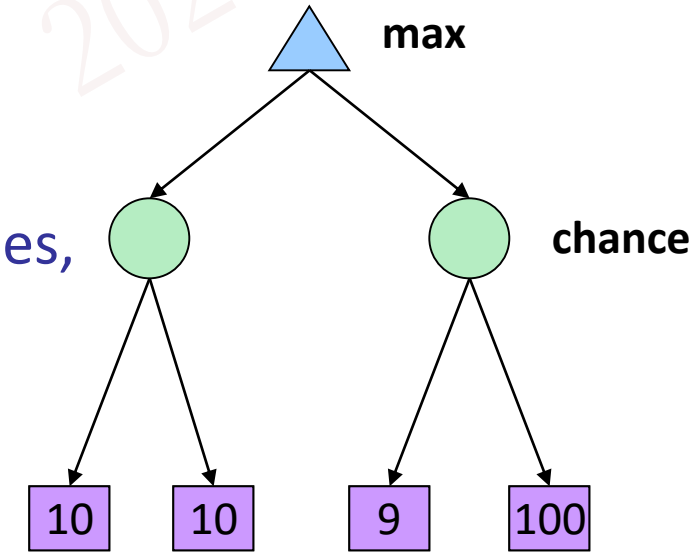


Uncertain outcomes controlled by chance, not an adversary!

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Expectimax Search

- Uncertain outcome
  - Explicit randomness: rolling dice
  - Unpredictable opponents: the ghosts respond randomly
  - Actions can fail: when moving a robot, wheels might slip
- Values should now reflect average-case (expectimax) outcomes, not worst-case (minimax) outcomes
- Expectimax search (**期望极大搜索**): compute the average score
  - Max nodes as in minimax search
  - Chance nodes (机会节点) are like min nodes but the outcome is uncertain
  - Calculate their expected utilities
  - I.e. take weighted average (expectation) of children nodes
- Later, we'll learn how to formalize the underlying uncertain-result problems as Markov Decision Processes (**马尔科夫决策过程**)



Expectimax Pseudocode

```
def value(state):
    if the state is a terminal state: return the state's utility
    if the next agent is MAX: return max-value(state)
    if the next agent is EXP: return exp-value(state)
```

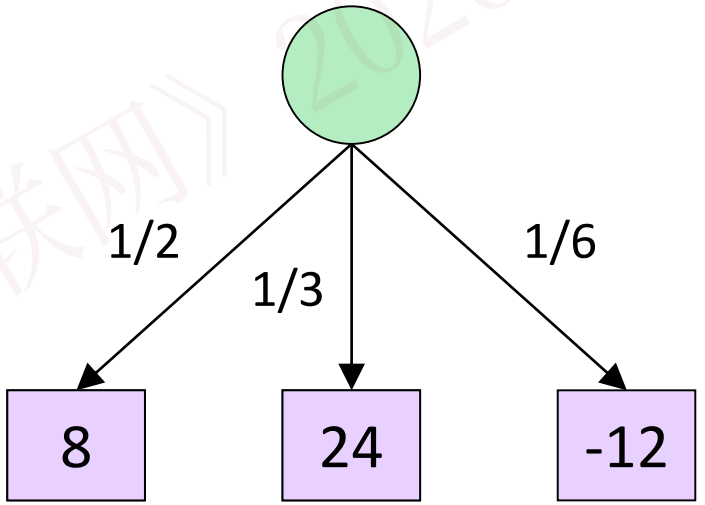
```
def max-value(state):
    initialize v = -∞
    for each successor of state:
        v = max(v, value(successor))
    return v
```

```
def exp-value(state):
    initialize v = 0
    for each successor of state:
        p = probability(successor)
        v += p * value(successor)
    return v
```

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Expectimax Pseudocode

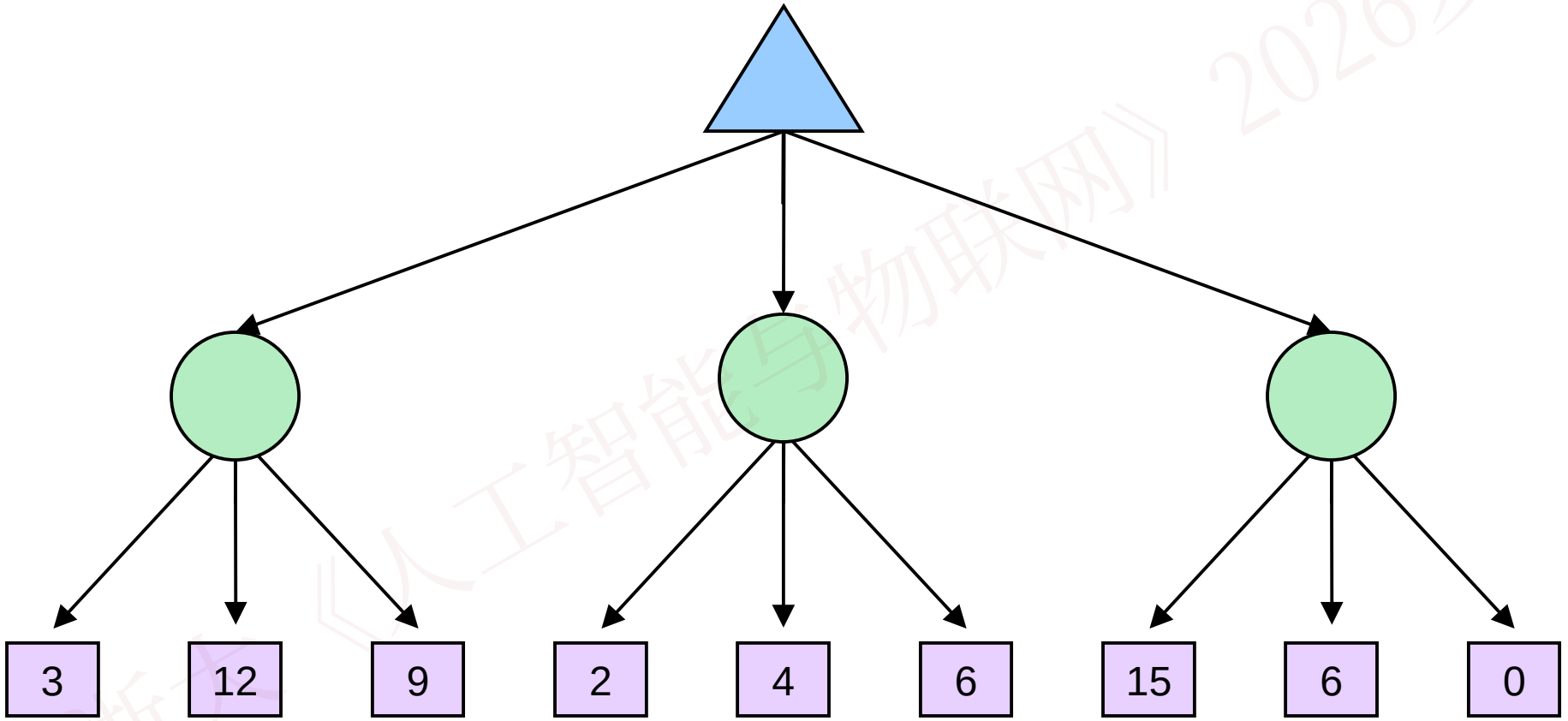
```
def exp-value(state):
    initialize v = 0
    for each successor of state:
        p = probability(successor)
        v += p * value(successor)
    return v
```



$$v = (1/2)\ (8) + (1/3)\ (24) + (1/6)\ (-12) = 10$$

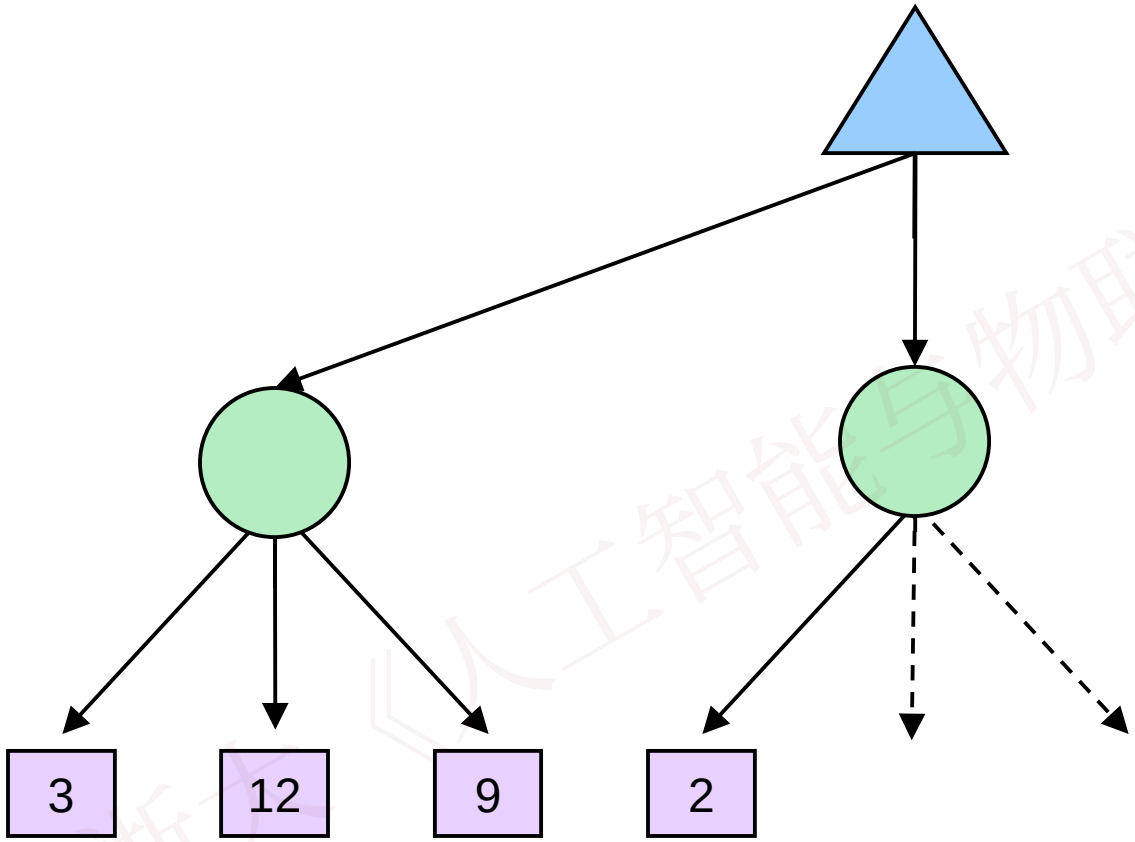
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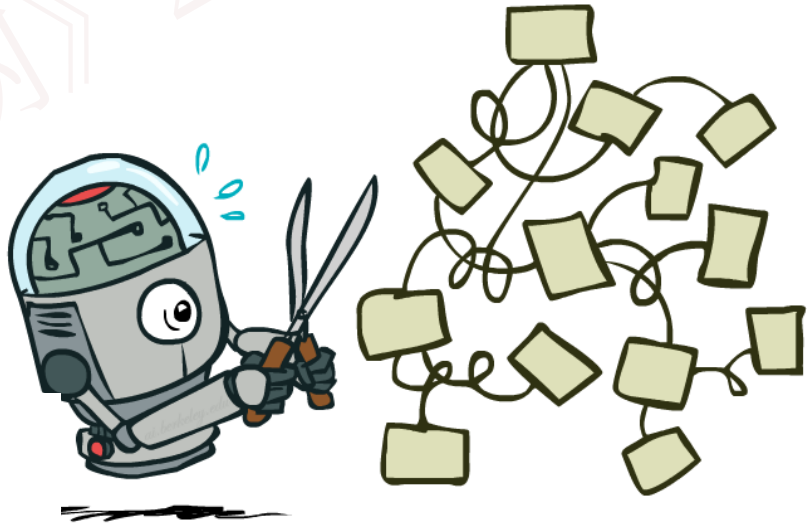
Expectimax Example



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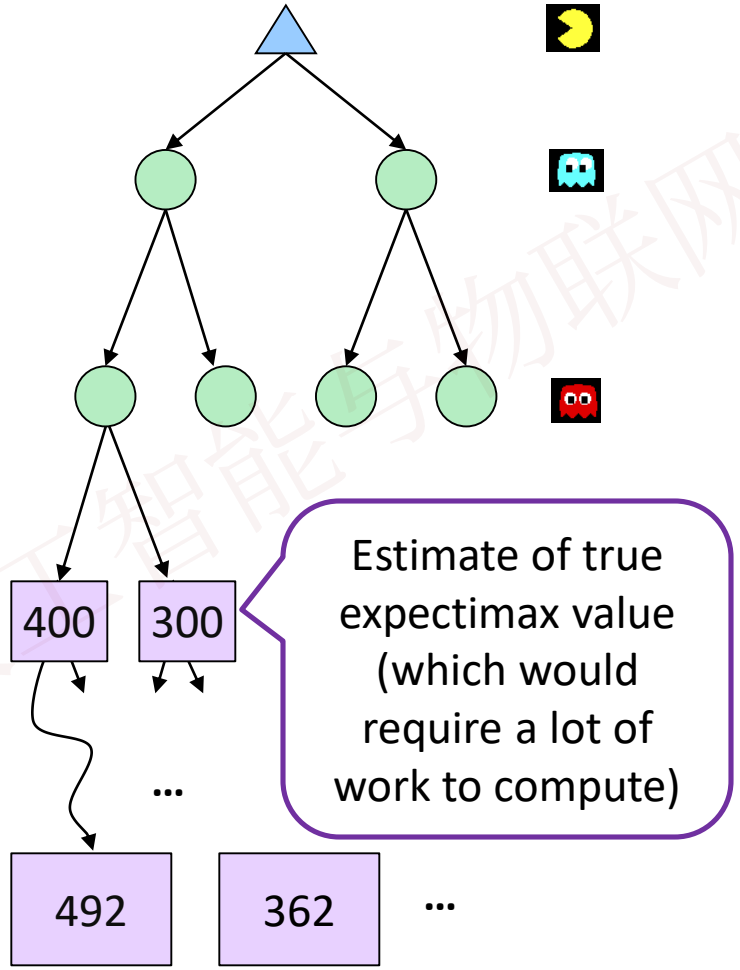
Expectimax Pruning?





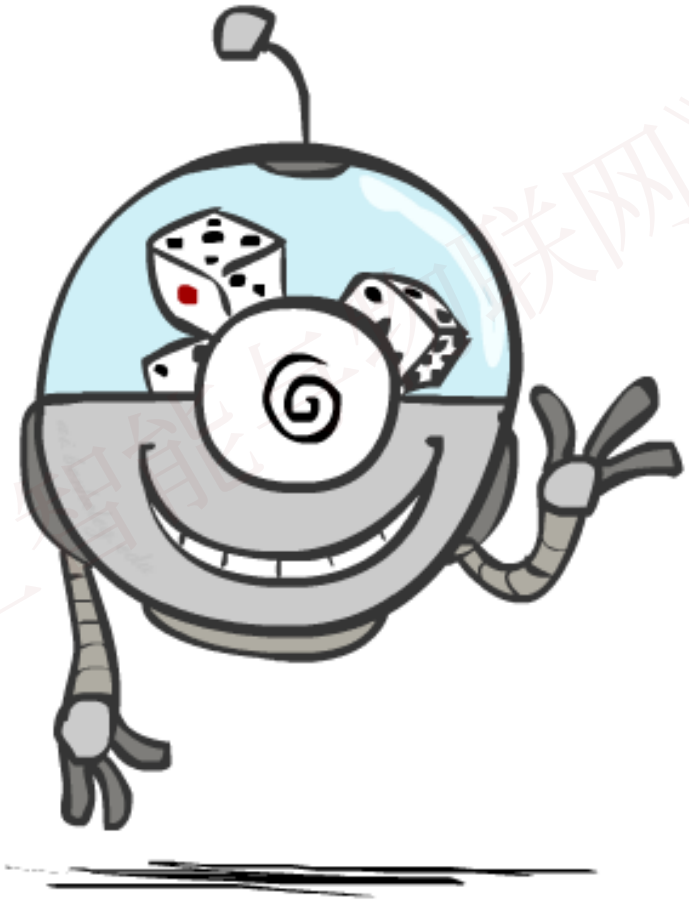
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Depth-Limited Expectimax



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Probabilities



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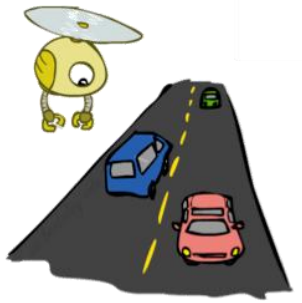
Reminder: Probabilities

- A random variable (随机变量) represents an event (事件) whose outcome (结果) is unknown
- A probability distribution (概率分布) is an assignment of weights to outcomes



0.25

- Example: Traffic on freeway
  - Random variable: T = whether there's traffic
  - Outcomes: T in {none, light, heavy}
  - Distribution: P(T=none) = 0.25, P(T=light) = 0.50, P(T=heavy) = 0.25



0.50

- Some laws of probability (more later):
  - Probabilities are always non-negative
  - Probabilities over all possible outcomes sum to one



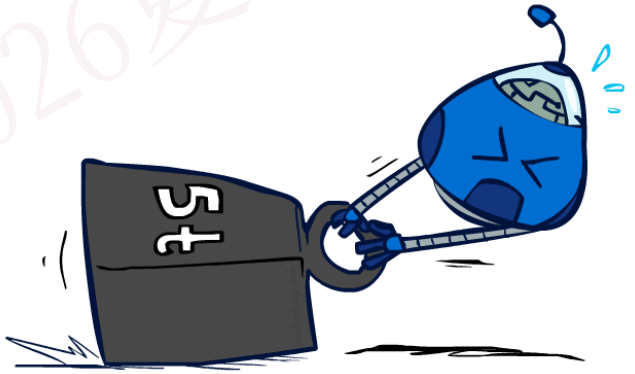
0.25

- As we get more evidence, probabilities may change:
  - P(T=heavy) = 0.25, P(T=heavy | Hour=8am) = 0.60
  - We'll talk about methods for reasoning and updating probabilities later

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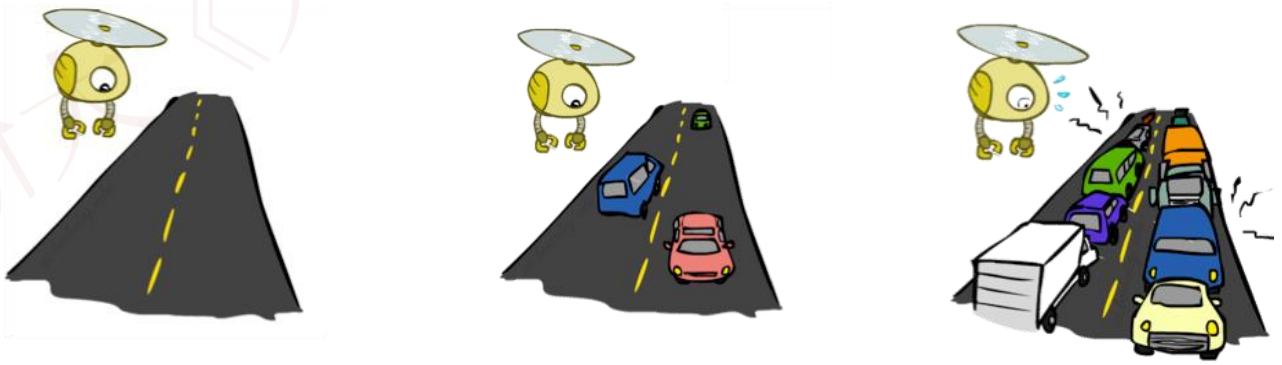
Reminder: Expectations

- The expected value of a function of a random variable is the average, weighted by the probability distribution over outcomes



- Example: How long to get to the airport?

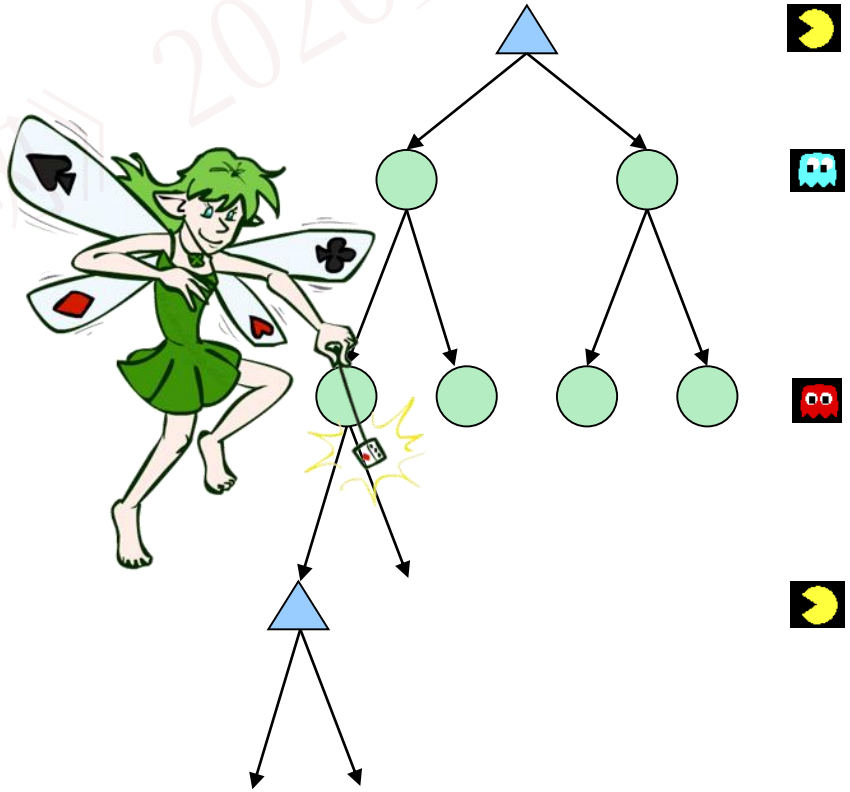
Time:	20 min		30 min		60 min		
	x	+	x	+	x		
Probability:	0.25		0.50		0.25		35 min



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What Probabilities to Use?

- In expectimax search, we have a probabilistic model of how the opponent (or environment) will behave in any state
  - Model could be a simple uniform distribution (roll a die)
  - Model could be sophisticated and require a great deal of computation
  - We have a chance node for any outcome out of our control: opponent or environment
- For now, assume each chance node magically comes along with probabilities that specify the distribution over its outcomes



Modeling Assumptions



The Dangers of Optimism and Pessimism

Dangerous Optimism

Assuming chance when the world is adversarial

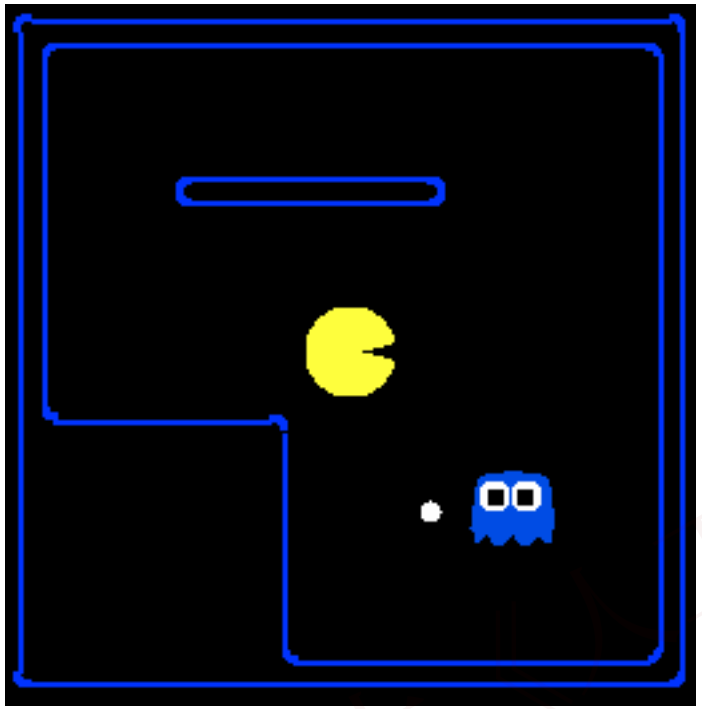


Dangerous Pessimism

Assuming the worst case when it's not likely



Assumptions vs. Reality



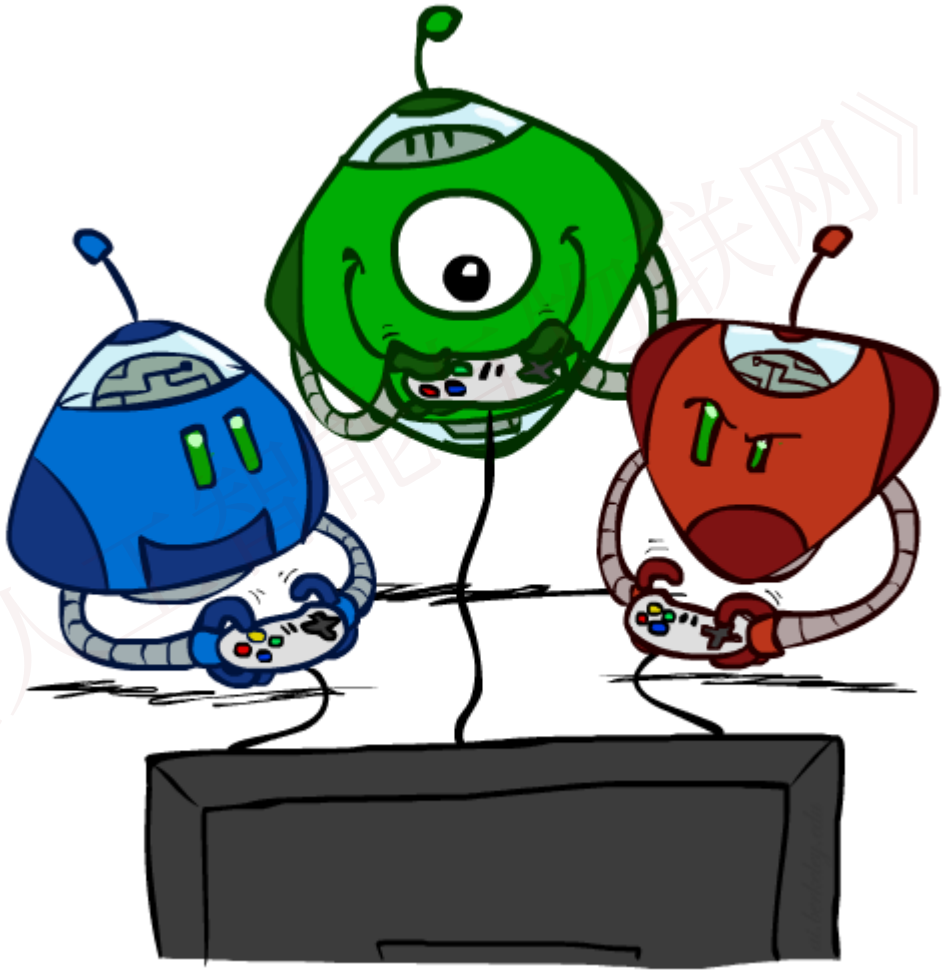
	Adversarial Ghost	Random Ghost
Minimax Pacman	Won 5/5 Avg. Score: 483	Won 5/5 Avg. Score: 493
Expectimax Pacman	Won 1/5 Avg. Score: -303	Won 5/5 Avg. Score: 503

Results from playing 5 games

Pacman used depth 4 search with an eval function that avoids trouble
Ghost used depth 2 search with an eval function that seeks Pacman

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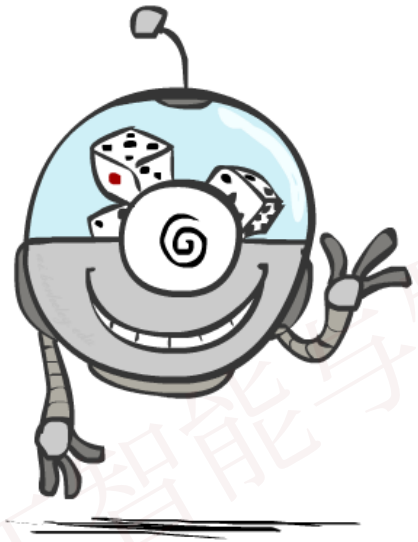
Other Game Types

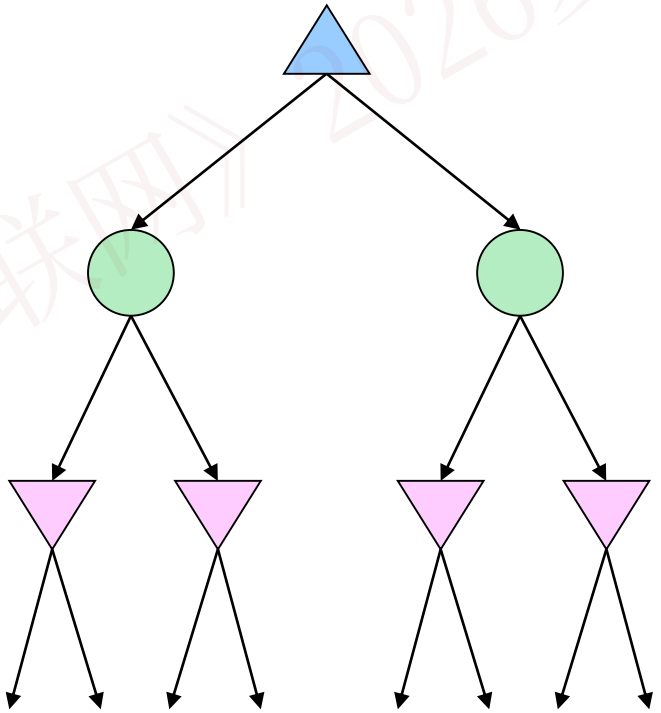


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Mixed Layer Types

- E.g. Backgammon
- Expectiminimax
  - Environment is an extra “random agent” player that moves after each min/max agent
  - Each node computes the appropriate combination of its children







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Example: Backgammon

- Dice rolls increase *b*: 21 possible rolls with 2 dice
  - Backgammon ≈ 20 legal moves

- Historic AI: TDGammon uses depth-2 search + very good evaluation function + reinforcement learning:
  world-champion level play

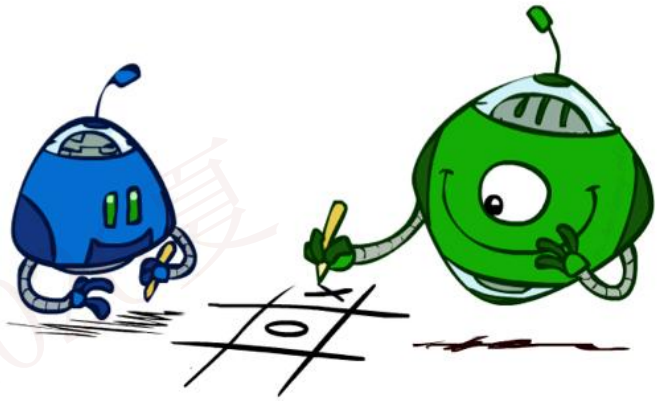
- 1st AI world champion in any game!



Image: Wikipedia

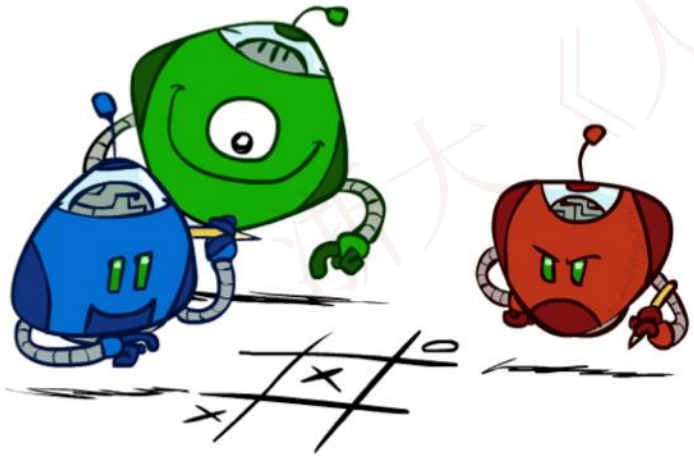
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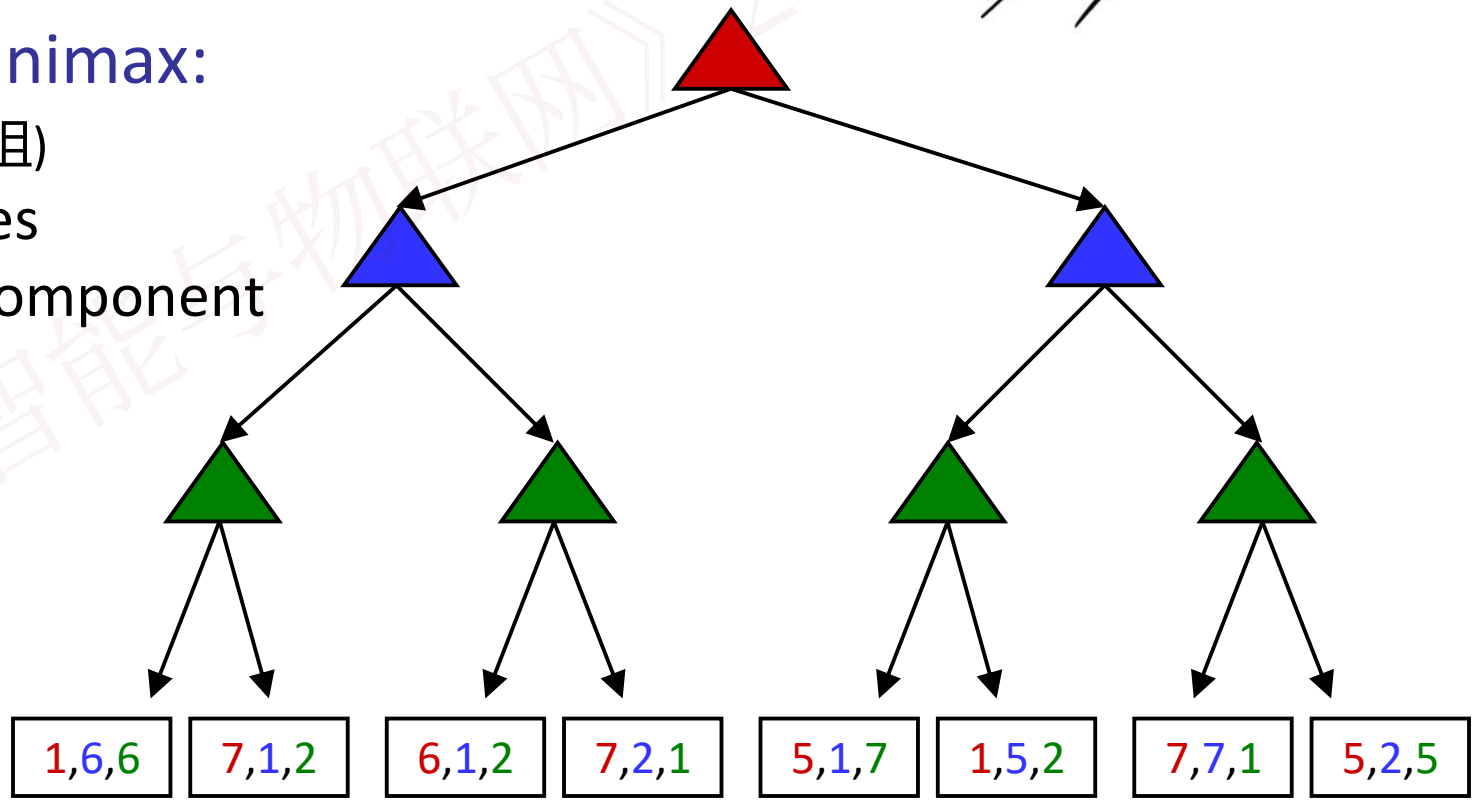
General Sum Games



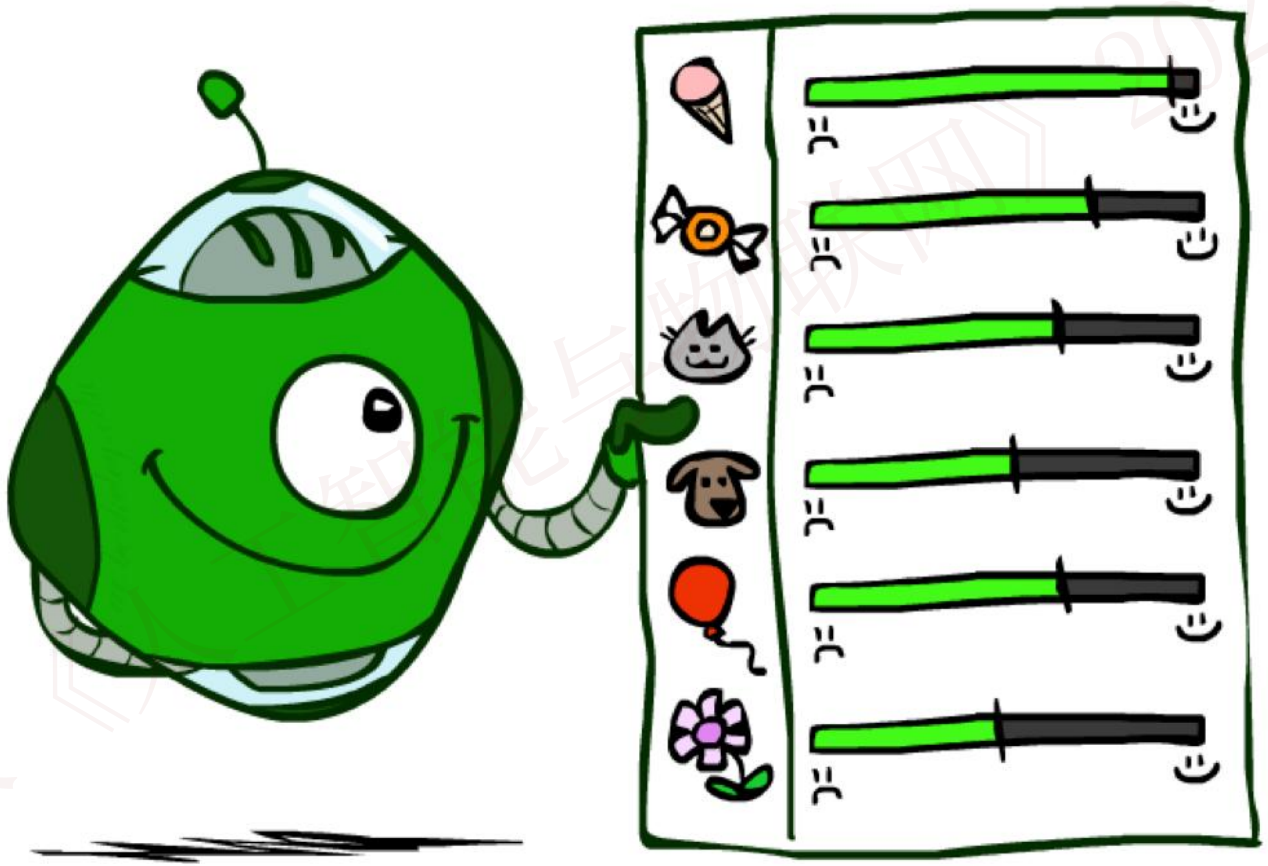
- What if the game is not zero-sum, or has multiple players?

- Generalization (可泛化性) of minimax:
  - Terminals have utility tuples (元组)
  - Node values are also utility tuples
  - Each player maximizes its own component
  - Can give rise to cooperation and competition dynamically…





Utilities



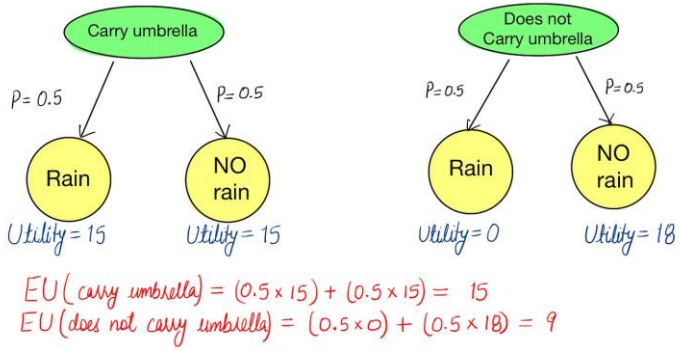
Maximum Expected Utility

- Why should we average utilities? Why not minimax?

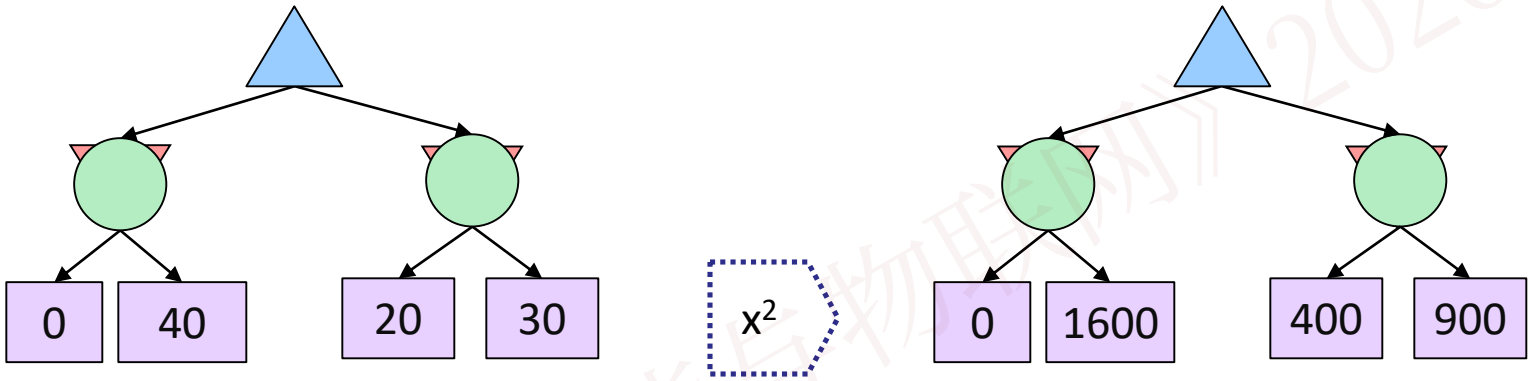
- Principle of maximum expected utility:
  - A rational agent should chose the action that maximizes its expected utility, given its knowledge



- Questions:
  - Where do utilities come from?
  - How do we know such utilities even exist?
  - How do we know that averaging even makes sense?
  - What if our behavior (preferences) can't be described by utilities?



What Utilities to Use?



- For worst-case minimax reasoning, terminal function scale doesn't matter
  - We just want better states to have higher evaluations (get the ordering right)
  - We call this insensitivity to monotonic transformations

- For average-case expectimax reasoning, we need *magnitudes* to be meaningful

Utilities

- Utilities are functions from outcomes (states of the world) to real numbers that describe an agent's preferences (偏好)

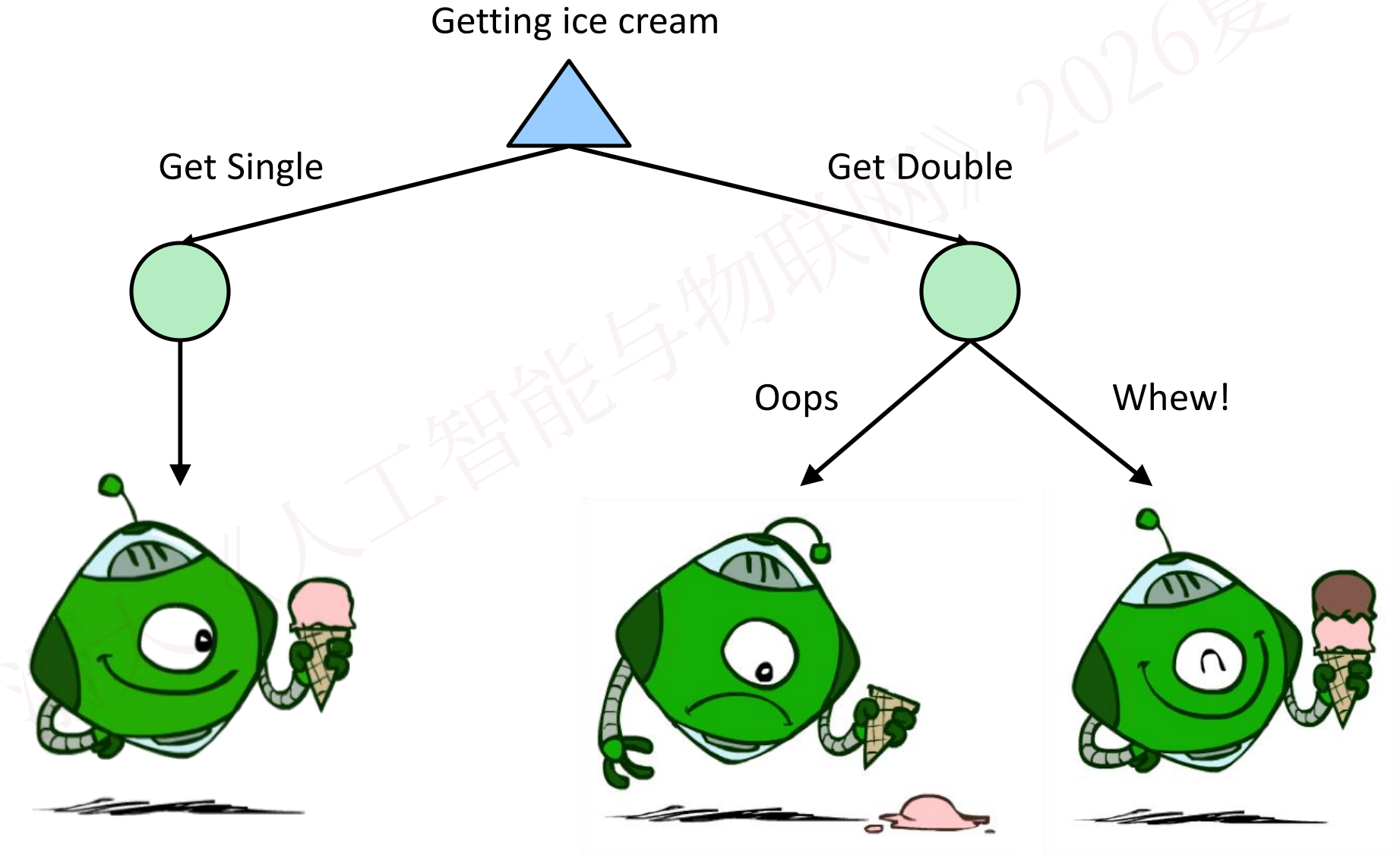
- Where do utilities come from?
  - In a game, may be simple (+1/-1)
  - Utilities summarize the agent's goals
  - Theorem: any "rational" preferences can be summarized as a utility function

- We define utilities and let behaviors emerge



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Utilities: Uncertain Outcomes



Preferences

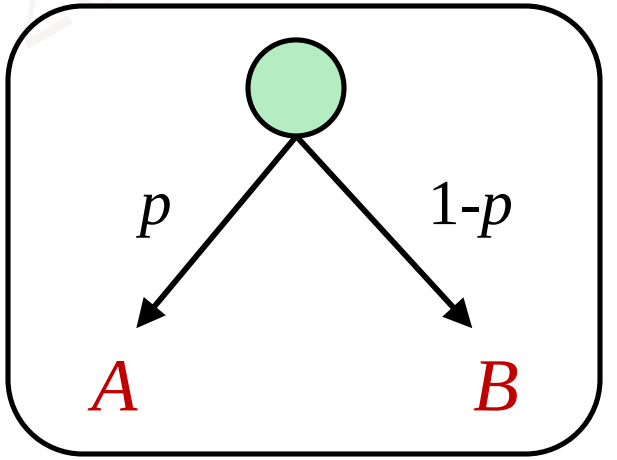
- An agent must have preferences among:
  - Prizes: A, B, etc.
  - Lotteries: situations with uncertain prizes

$$L = [p, A;\ (1-p), B]$$

A Prize



A Lottery

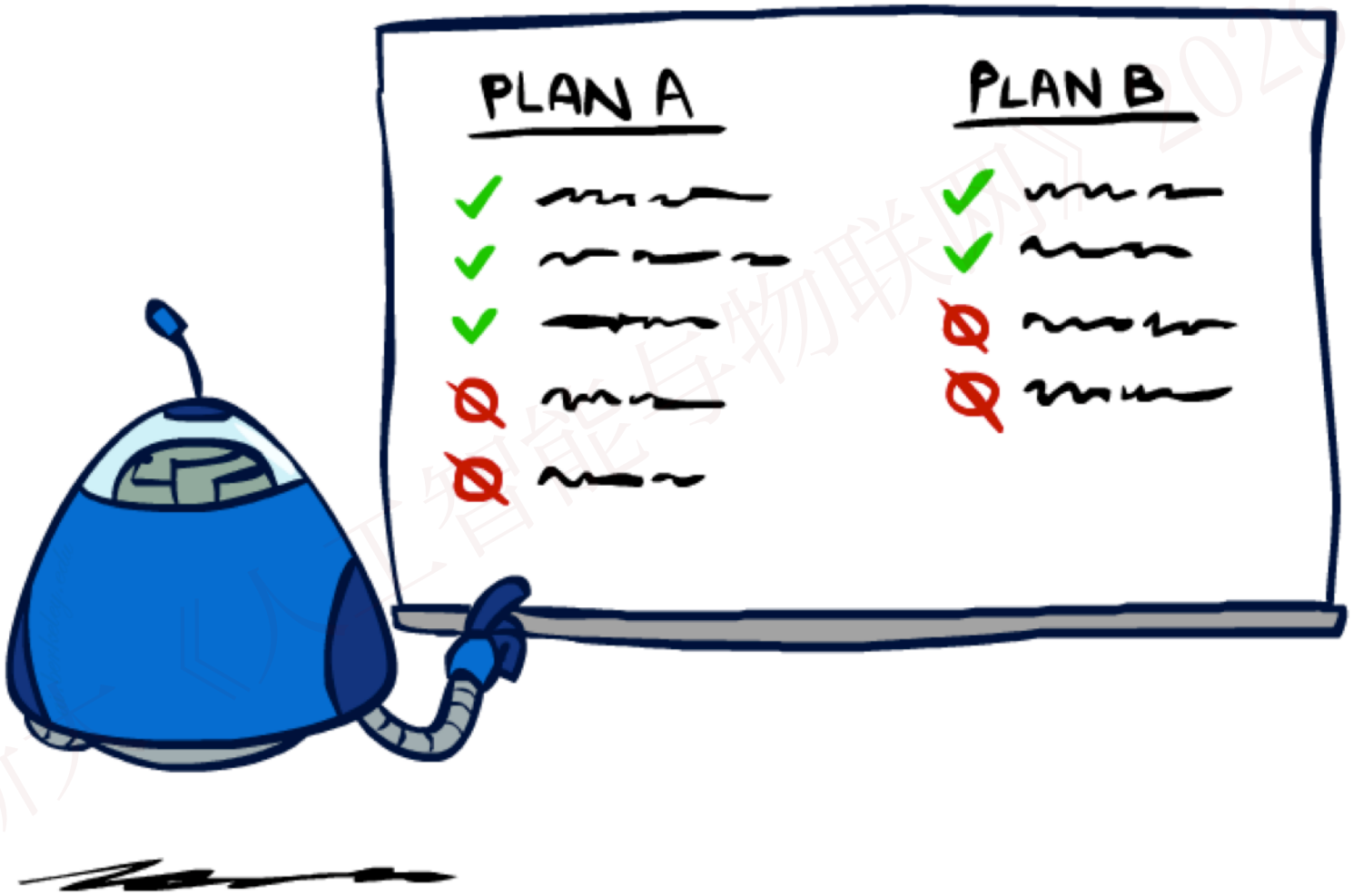


- Notation:
  - Preference: $A \succ B$
  - Indifference (无差别): $A \sim B$





Rationality (理性)

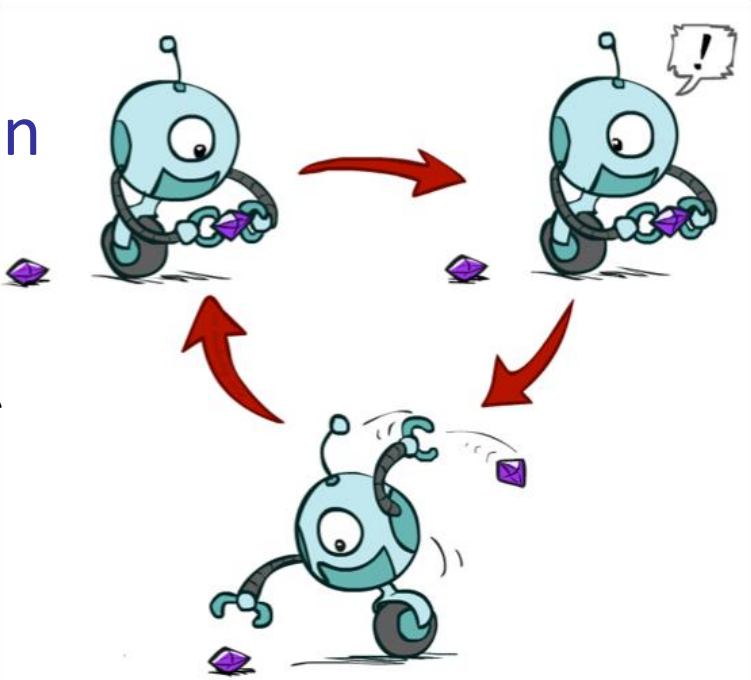


Rational Preferences

- We want some constraints on preferences before we call them rational, such as:

Axiom of Transitivity: $(A \succ B) \wedge (B \succ C) \Rightarrow (A \succ C)$

- For example: an agent with intransitive preferences can be induced to give away all of its money
  - If B > C, then an agent with C would pay (say) 1 cent to get B
  - If A > B, then an agent with B would pay (say) 1 cent to get A
  - If C > A, then an agent with A would pay (say) 1 cent to get C



Rational Preferences

The Axioms (公理) of Rationality

Orderability 可排序性

$$(A \succ B) \vee (B \succ A) \vee (A \sim B)$$

Transitivity 传递性

$$(A \succ B) \wedge (B \succ C) \Rightarrow (A \succ C)$$

Continuity 连续性

$$A \succ B \succ C \Rightarrow \exists p\ [p, A;\ 1-p, C] \sim B$$

Substitutability 可替代性

$$A \sim B \Rightarrow [p, A;\ 1-p, C] \sim [p, B; 1-p, C]$$

Monotonicity 单调性

$$A \succ B \Rightarrow$$
$$(p \geq q \Leftrightarrow [p, A;\ 1-p, B] \succeq [q, A;\ 1-q, B])$$



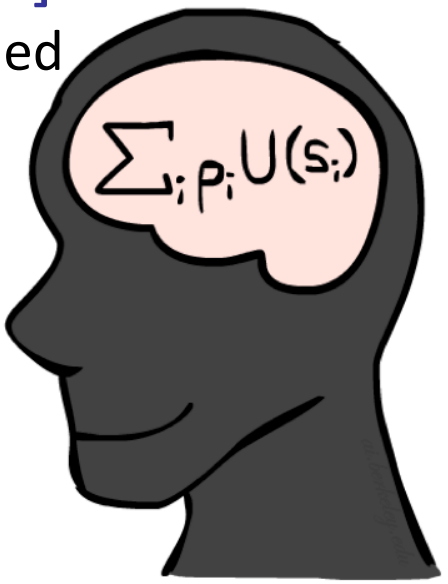
Theorem (定理): Rational preferences imply behavior describable as maximization of expected utility

Rational Preferences

- Theorem [Ramsey, 1931; von Neumann & Morgenstern, 1944]
  - Given any preferences satisfying these constraints, there exists a real-valued function U such that:

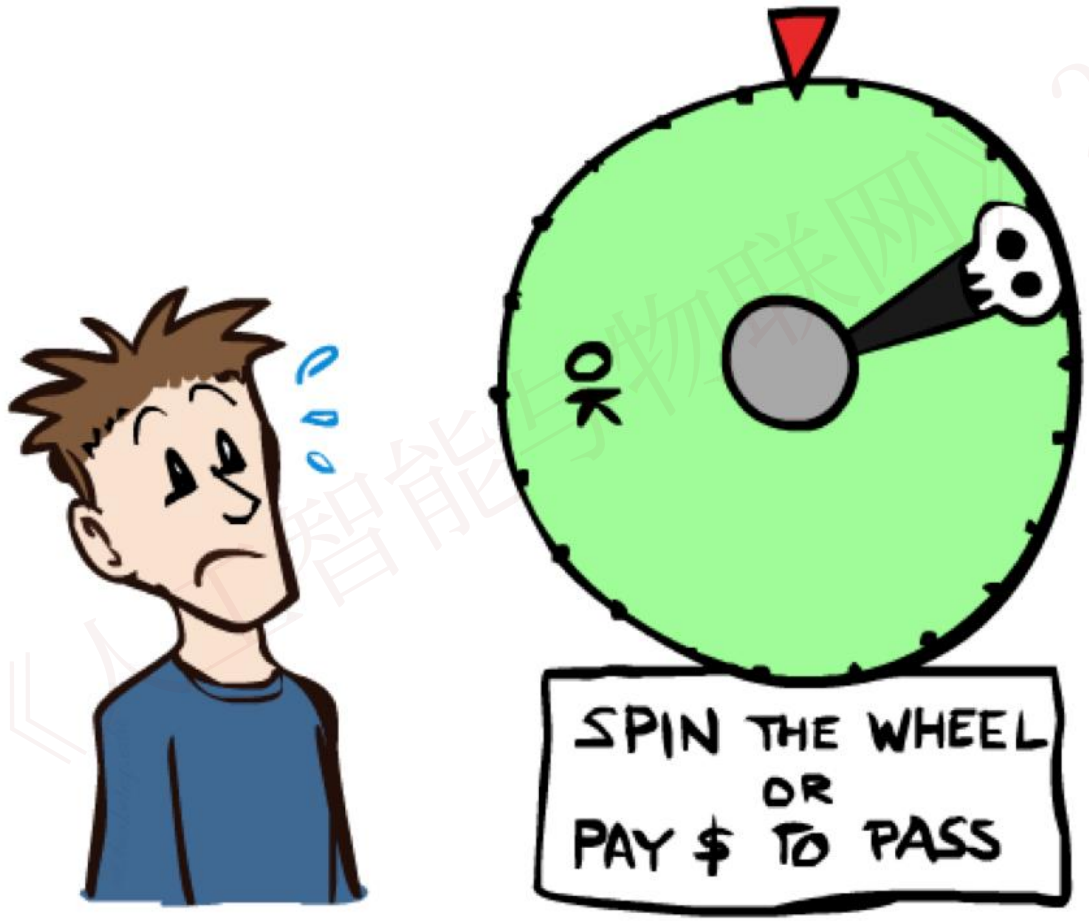
$$U(A) \geq U(B) \;\Leftrightarrow\; A \succeq B$$

$$U([p_1, S_1;\; \ldots\; ;\; p_n, S_n]) = \sum_i p_i U(S_i)$$



- Maximum expected utility (MEU) principle:
  - Choose the action that maximizes expected utility
  - Note: an agent can be entirely rational (consistent with MEU) without ever representing or manipulating utilities and probabilities
  - E.g., a lookup table for perfect tic-tac-toe, a reflex vacuum cleaner

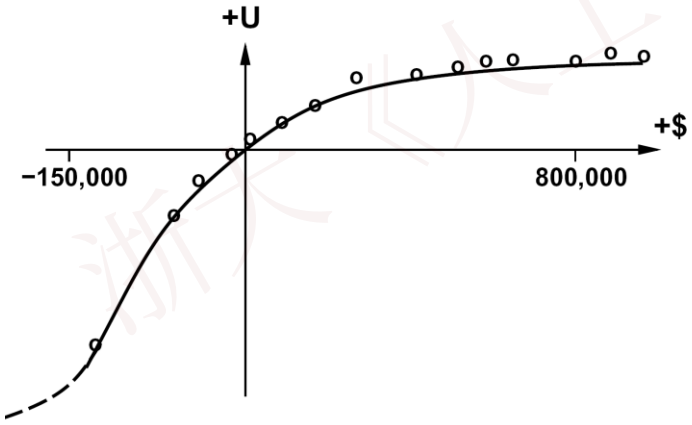
Human Utilities



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Money

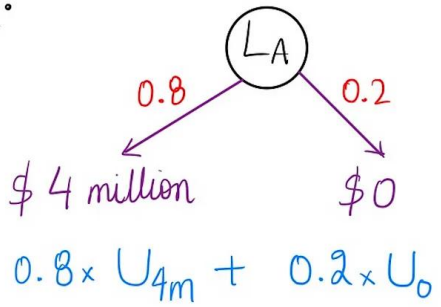
- Money does not behave as a utility function, but we can talk about the utility of having money (or being in debt)
- Given a lottery L = [p, $X; (1-p), $Y]
  - The expected monetary value EMV(L) is p*X + (1-p)*Y
  - U(L) = p*U($X) + (1-p)*U($Y)
  - Typically, U(L) ≠ U(EMV(L))
  - In this sense, people are risk-averse (风险厌恶)

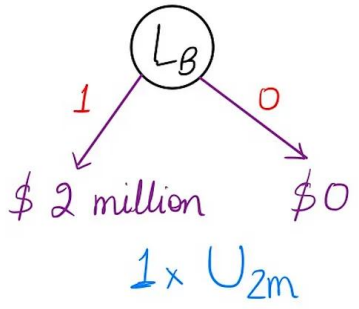






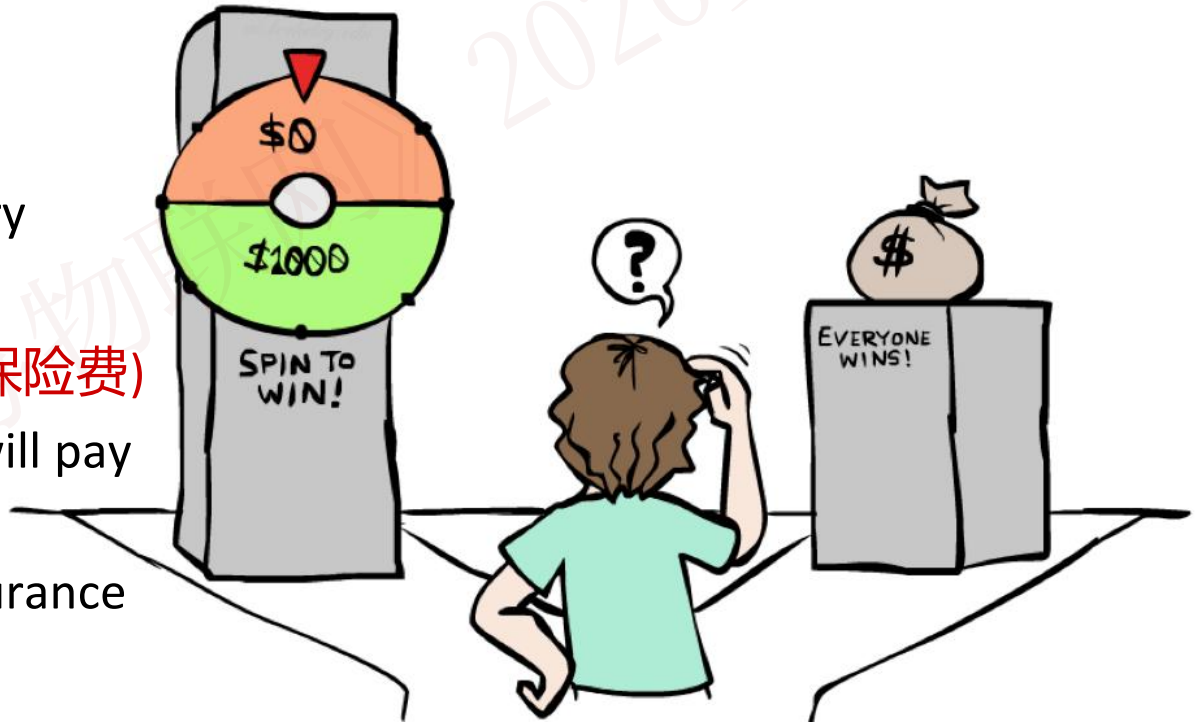
Case 2:





Example: Insurance

- Consider the lottery [0.5, $1000; 0.5, $0]
  - What is its expected monetary value? ($500)
  - What is its certainty equivalent?
    - Monetary value acceptable to replace the lottery
    - $400 for most people
  - Difference of $100 is the insurance premium (保险费)
    - There's an insurance industry because people will pay to reduce their risk
    - If everyone were risk-neutral (风险中性), no insurance needed!
  - It's win-win: you'd rather have the $400 and the insurance company would rather have the lottery (they have many lotteries)



Example: Human Rationality?

- Famous example of Allais (1953)
  - A: 0.8 win $8,000.
  - B: 1.0 win $6,000.
  - C: 0.02 win $8,000.
  - D: 0.025 win $6,000.
- Most people prefer B > A, C > D
- But if U($0) = 0, then
  - B > A $\Rightarrow$ U($6k) > 0.8 U($8k)
  - C > D $\Rightarrow$ 0.8 U($8k) > U($6k)

